ModelIt!

Teacher's Guide

Can You Predict the Weather?

Patterns in Weather and Climate

Quick Reference

Grade Level	Grade 3
NGSS Standard	3-ESS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 3-ESS2-1

Represent data in tables and graphical displays to describe typical weather conditions expected during a particular season.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students analyze weather data displayed in tables and graphs to identify seasonal patterns and make evidence-based weather predictions.
Disciplinary Core Idea	**ESS2.D Weather and Climate** — Scientists record patterns of the weather across different times and areas so that they can make predictions about what kind of weather might happen next.
Crosscutting Concept	**Patterns** — Students identify repeating patterns in weather data across seasons and use those patterns to predict future weather conditions.

CAST Testing Alignment

MULTIPLE CHOICE:

A student collects temperature data for one year. She notices that July is always the hottest month. Which best describes what she discovered?

CONSTRUCTED RESPONSE:

Your friend says the weather is totally random and nobody can predict it. Using what you know about weather patterns, explain why meteorologists CAN predict the weather. Use the words 'pattern' and 'data' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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> **For Your Confidence**
>
> This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Weather affects our lives every single day — what we wear, what we do outside, and even how we feel. Understanding weather patterns helps us plan ahead and stay safe during storms, heat waves, and cold snaps.

What Is Weather?

Weather describes the conditions of the atmosphere at a specific time and place. It includes temperature (how hot or cold), precipitation (rain, snow, sleet, hail), wind (how fast and which direction air moves), humidity (how much moisture is in the air), and cloud cover (how many clouds are in the sky). Weather can change quickly — it might be sunny in the morning and stormy by afternoon.

What Is Climate?

Climate is the average pattern of weather in a place over 30 or more years. While weather changes day to day, climate changes very slowly. For example, the weather in Phoenix, Arizona on any given day might be cloudy, but the climate there is hot and dry. Knowing the climate helps us predict what KINDS of weather to expect in a place during each season.

How Do Meteorologists Predict Weather?

Meteorologists are scientists who study weather. They use tools like thermometers (temperature), barometers (air pressure), anemometers (wind speed), and rain gauges (precipitation). They also use satellites, weather balloons, and radar. By collecting data over time, they spot patterns. They know that certain cloud types bring rain, dropping air pressure often means a storm is coming, and seasons follow predictable temperature patterns. Modern computers help them make forecasts up to 10 days ahead.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify temperature as an external component and cloud cover and chance of rain as internal components that change in response to temperature conditions.

E **Establish Relationships**
Students determine that temperature affects cloud cover and chance of rain — warmer air holds more moisture, and different temperatures create different weather patterns.

V **Visualize & Model**
Students build a model showing how temperature connects to cloud cover and chance of rain in a weather system.

E **Evaluate Outcomes**
Students run hot summer day, cold winter day, and cool spring day scenarios to observe how changing temperature affects cloud cover and precipitation.

R **Revise & Extend**
Students add wind speed or humidity to explore how additional factors make weather prediction more accurate but also more complex.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look outside right now — what is the weather doing? Could you have guessed this morning what it would be like?"*

Watch For

Have students look out the window and describe today's weather. Ask: Do you think we could have predicted this yesterday?

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to become weather detectives and figure out if we can predict what the sky will do next!"*

Watch For

Read objectives together. Introduce vocabulary — have students act out different types of weather (shivering for cold, fanning for hot).

Slide 3: The Big Question

3 min

Say: *"Can we actually predict the weather? Or is it just random? Let us investigate!"*

Watch For

Show a week of weather data from your city. Ask students: Do you see any patterns? Turn and talk with a partner.

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a weather model to see if patterns can help us predict rain and sunshine!"*

Watch For

Preview each LEVER step. Explain that weather scientists use models every day to make forecasts.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What parts of weather can we measure? Which one do we set, and which ones change on their own?"*

Watch For

Guide sorting of three components. Explain temperature is external because it is set by the sun and seasons.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When the temperature goes up in summer, what usually happens to clouds and rain? Let us connect the parts!"*

Watch For

Help students draw arrows showing how temperature connects to cloud cover and chance of rain.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Let us test our weather model! We will try a hot summer day, a cold winter day, and a cool spring day."*

Watch For

Have students predict first, then run each scenario. Compare results across seasons on the graph.

Slide 8: What Did We Discover?

7 min

Say: *"Now we know the secret — weather follows PATTERNS, and patterns help us predict what comes next!"*

Watch For

Summarize discoveries. Show a real 7-day forecast and point out the patterns. Ask: What do you predict for next week?

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Temperature
- **INTERNAL:** Cloud Cover, Chance of Rain

Relationships

- **Temperature to Cloud Cover:** POSITIVE (+) — When temperature rises, more water evaporates into the air, which can form more clouds. Warm, humid days often have big puffy clouds building up, especially in summer afternoon thunderstorms.
- **Cloud Cover to Chance of Rain:** POSITIVE (+) — More clouds in the sky means more water droplets are gathered together. When enough water collects in clouds, it falls as precipitation. A sky full of dark clouds has a much higher chance of rain than a clear blue sky.

Key Discoveries

- Weather follows patterns — certain seasons tend to have similar types of weather year after year
- Temperature, cloud cover, and precipitation are all connected in a weather system
- Scientists called meteorologists use patterns in weather data to make predictions about future weather
- Climate is the long-term pattern of weather — it tells us what weather is TYPICAL for a place and season

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Weather is completely random and cannot be predicted"	Weather follows patterns that repeat over time, especially with the seasons. Meteorologists use data, patterns, and computer models to make predictions. Forecasts are not perfect, but they are right most of the time because weather is NOT random.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Climate and weather are the same thing"	Weather is what is happening right now — it changes every day. Climate is the average pattern of weather over 30+ years. You can have a cold day in summer (weather) even though summers are usually hot (climate). Climate is like your personality — weather is like your mood today.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Hot weather always means sunny and no rain"	Some of the rainiest places on Earth are also the hottest, like tropical rainforests. Hot air can hold more moisture, which can lead to bigger storms. Summer thunderstorms happen BECAUSE it is hot — the heat causes air to rise rapidly, forming towering storm clouds.